15 Ali and Sam are playing a game in which Ali tosses a coin 5 times. If there are 4 or 5 heads, Ali wins the game. Otherwise Sam wins the game. They decide to play the game 50 times.

(a) Initially Sam models the situation by assuming the coin is fair. Determine the number of games **Ali** is expected to win according to this model. **[2]**

Ali thinks the coin may be biased, with probability p of obtaining heads when the coin is tossed. Before playing the game, Ali and Sam decide to collect some data to estimate the value of p. Sam tosses the coin 15 times and records the number of heads obtained. Ali tosses the coin 25 times and records the number of heads obtained.

(b) Explain why it is better to use the combined data rather than just Sam's data or just Ali's data to estimate the value of p. **[1]**

Ali records 20 heads and Sam records 8 heads.

(c) Use the combined data to estimate the value of p. **[1]**

Ali now models the situation using the value of p found in part **(c)** as the probability of obtaining heads when the coin is tossed.

(d) Determine how many games Ali expects to win using this value of p to model the situation. **[2]**

(e) Ali wins 25 of the 50 games. Explain whether Sam's model or Ali's model is a better fit for the data. **[1]**